INTELLI-CREDIT

AI-Powered Credit Decisioning Engine

Hackathon Pitch Preparation Document

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Team SPB

1. Problem Statement

The Challenge: In Indian corporate lending, credit managers face a 'Data Paradox' - overwhelmed by data yet taking **weeks** to process a single loan application.

Data sources that must be stitched together:

- **Structured:** GST filings, ITRs, Bank Statements
- **Unstructured:** Annual Reports, Financial Statements, Board Minutes, Rating Reports
- **External:** News, MCA filings, Legal disputes (e-Courts)
- **Primary:** Factory site visits, Management interviews

What Was Asked: Build an AI-powered Credit Decisioning Engine that automates end-to-end preparation of a Comprehensive Credit Appraisal Memo (CAM).

Pillar	Requirement	Our Module(s)
1. Data Ingestor	Multi-format parsing, GST vs Bank cross-verification	data_ingestor.py, claude_pdf_processor.py, ai_analyzer.py, cross_verifier.py
2. Research Agent	Web crawling, news/legal/regulatory, qualitative input portal	research_agent.py, nlp_engine.py
3. Recommendation Engine	CAM generation, explainable scoring, loan amount & rate	scoring_engine.py, credit_analyzer.py, credit_risk_model.py, cam_generator.py

Evaluation Criteria:

- **Extraction Accuracy** - How well we extract from messy, scanned Indian PDFs
- **Research Depth** - Finding relevant local news and regulatory filings
- **Explainability** - Can AI walk the judge through its logic
- **Indian Context Sensitivity** - GSTR-2A vs 3B, NBFC nuances, CIBIL

2. Proposed Solution & Methodology

The 8-Step Pipeline

Step	Module	What It Does
1. Data Ingestion	data_ingestor.py	Parses PDFs (PyMuPDF + pdfplumber + OCR), Excel/CSV
2. AI Extraction	claude_pdf_processor.py + ai_analyzer.py	Extracts 30+ financial metrics using Claude + regex
3. Cross-Verification	cross_verifier.py	GST vs Bank reconciliation, circular trading detection
4. Research Agent	research_agent.py	Web crawls for news, legal, MCA filings, regulatory actions
5. NLP Analysis	nlp_engine.py	Sentiment analysis, risk signals, entity recognition
6. ML Prediction	credit_risk_model.py	XGBoost classifier + SHAP explainability
7. Credit Scoring	credit_analyzer.py + scoring_engine.py	Entity classification, 15 ratios, Five Cs scoring
8. CAM Generation	cam_generator.py	Professional Word/PDF Credit Appraisal Memo

Dual-Track Extraction (Unique Approach)

Track A - Claude API: Sends full PDF to Claude 3.5 Sonnet with structured JSON schema for 30+ metrics. Handles complex layouts end-to-end.

Track B - Regex Engine: 10 specialized extraction functions with unit detection, bracket-negative handling. Zero API cost, deterministic, reproducible.

Result: Best of both tracks merged - Claude for accuracy, Regex for reliability.

3. Calculations Performed

3.1 Financial Metric Extraction

Metric	Method
Revenue	Regex: 'Total revenue from operations' from P&L;
PAT	Regex: 'Owners of the company' from Consolidated P&L; (NOT auditor's report)
Net Worth	Regex: 'Equity attributable to owners' from BS; fallback: Total Assets - Total Debt
Total Debt	Component sum: Debt Securities + Borrowings + Subordinated Liabilities
EBITDA	Direct extraction; NBFCs: PPOP = Revenue - Costs
Interest Expense	From P&L; Expenses section: 'Finance costs'
Operating CF	'Net cash from operating activities' with bracket-negative handling
NII, GNPA, NNPA	Percentage extraction for NBFC-specific metrics

3.2 Computed Credit Ratios (15 Ratios)

Ratio	Formula	Benchmark
D/E Ratio	Total Debt / Net Worth	< 2.0
ICR	EBITDA / Interest Expense	> 3.0
DSCR	(PAT + Depr + Interest) / (Interest + Principal)	> 1.5
Current Ratio	Current Assets / Current Liabilities	> 1.5
EBITDA Margin	EBITDA / Revenue x 100	Industry-specific
PAT Margin	PAT / Revenue x 100	Higher is better
Revenue Growth	(Curr - Prior) / Prior x 100	Always algebraic
ROA	PAT / Avg Total Assets x 100	> 2% for NBFCs
ROE	PAT / Avg Net Worth x 100	> 15%
GNPA Ratio	Gross NPA / Advances x 100	< 3%
CRAR/CAR	Capital Adequacy Ratio	> 15% (RBI)

3.3 Cross-Verification Checks

Check	Method
Revenue Reconciliation	GST revenue vs bank credits; flag if > 10% variance
Circular Trading	Matching credit-debit amounts > 1L in bank statements
Revenue Inflation	Coefficient of variation on monthly GST revenue; flag if CV > 0.5
Cash Flow Consistency	Inflows vs outflows ratio analysis
GSTR-2A vs 3B Mismatch	ITC claimed vs auto-populated; flag if > 10% variance

3.4 Five Cs Scoring Model

C	Weight	Key Inputs
Character	25%	Media sentiment, risk signals, legal issues, promoter reputation
Capacity	25%	DSCR, ICR, revenue growth, PAT margin, operating cash flow
Capital	20%	D/E ratio, net worth, total assets, ROE/ROA
Collateral	15%	Current ratio, asset coverage, tangible assets
Conditions	15%	Industry outlook, regulatory risk, sector growth

Final Score = Sum(C_score x Weight) - CV Penalty +/- Qualitative Adjustment

4. AI/ML Models Used & Why

4.1 Models We Use

Model	Purpose	Why This Model
Claude 3.5 Sonnet	PDF extraction	Best document understanding; handles messy Indian PDFs; 200K context
GPT-4o	Narrative/CAM	Superior narrative quality; understands Indian financial context
XGBoost	Risk classification	Industry-standard for tabular data; handles missing values; fast
SHAP	Explainability	Feature-level explanations for every prediction
Custom NLP Lexicon	Sentiment analysis	100+ Indian credit keywords; faster than LLM for batch
Regex Engine	Data extraction	Unit-aware, bracket-negative, multi-column year extraction

4.2 Why NOT These Models

Model	Why Not
FinBERT/BERT	Overkill; our custom lexicon is faster and Indian-context specific
Random Forest	XGBoost consistently outperforms on tabular financial data
LLaMA/Open-source	Needs GPU infrastructure; latency too high; weaker at extraction
Tesseract only	PyMuPDF + pdfplumber handle native PDFs better; Tesseract is fallback

5. How Our Model Is Different

Feature	Traditional Tools	IntelliCredit
Processing Time	2-3 weeks	< 5 minutes
Data Sources	Manual collection	Automated multi-source
Fraud Detection	Post-facto audits	Real-time circular trading + GST mismatch
NBFC Support	One-size-fits-all	Dedicated NBFC pipeline (PPOP, NII, GNPA, CRAR)
Explainability	Black box scores	SHAP + Five Cs + factor-level reasoning
Research	Manual Google	Automated web crawling
CAM Generation	Manual Word doc	Auto-generated professional Word/PDF
Indian Context	Western models	Built for India (Lakhs/Crores, GSTR, NCLT)

Indian Context Sensitivity Features

- **Unit Handling:** Auto-detects Lakhs/Crores/Millions from PDF headers; normalizes to Crores
- **GSTR-2A vs 3B:** Specific cross-verification for ITC mismatch detection
- **NBFC Classification:** Auto-detects from keywords (NII, GNPA, AUM, CRAR); adapts all metrics
- **Indian Accounting:** Handles bracket notation for negatives: (22,489.55) = -22,489.55
- **Credit Ratings:** Extracts CRISIL, ICRA, CARE, India Ratings, Brickwork, Acuite
- **Legal Context:** Monitors NCLT, SFIO, CBI, ED, SEBI, RBI actions
- **10+ Sanity Checks:** PAT < Revenue, GNPA > NNPA, ROA != ROE, CRAR > 10%

6. Implementation Feasibility & Impact

Technical Stack

Layer	Technology
Backend	Python 3.x, Flask
AI/LLM	Claude 3.5 Sonnet (Anthropic), GPT-4o (OpenAI)
ML	XGBoost, SHAP, scikit-learn, NumPy, Pandas
PDF Processing	PyMuPDF (fitz), pdfplumber, Tesseract OCR
NLP	Custom Indian credit lexicon (100+ keywords)
Web Scraping	DuckDuckGo API, BeautifulSoup, Requests
Document Gen	python-docx (Word), ReportLab (PDF)
Frontend	HTML5, JavaScript, Chart.js
Data Integration	Databricks connector ready

Impact Metrics

Metric	Current (Manual)	With IntelliCredit
Processing Time	2-3 weeks	< 5 minutes
Cost per Application	Rs 15,000-25,000	Rs 5-10 (API costs)
Fraud Detection	Post-facto	Real-time
Consistency	Human-dependent	100% standardized
Coverage	8-10 data points	30+ metrics + web research
Explainability	Subjective notes	SHAP + Five Cs + Factors

Code Architecture: 6,500+ lines across 12 core modules

7. Pitch Talking Points

Opening (30 seconds)

> *"Credit managers spend 2-3 weeks per loan application stitching together data from 10+ sources. IntelliCredit does it in under 5 minutes with AI - and catches fraud that humans miss."*

Suggested Demo Flow

1. Upload an NBFC Annual Report PDF - show real-time extraction
2. Point out Net Worth, PAT, Revenue auto-extracted with unit normalization
3. Show Cross-Verification tab - circular trading detection in action
4. Show Research tab - live web-crawled news and legal findings
5. Show Credit Analysis - Five Cs breakdown with SHAP explanation
6. Download CAM - professional 10-section document auto-generated

Key Differentiators to Emphasize

- **Dual-track extraction** (Claude + Regex) - accuracy + reliability
- **NBFC-aware** - not just generic corporate analysis
- **Fraud detection** built into the pipeline (circular trading, GST mismatch)
- **Explainable** - every decision has SHAP + Five Cs + factor-level reasoning
- **Indian context** - Lakhs/Crores, GSTR, NCLT, bracket negatives
- **End-to-end** - from PDF upload to CAM download in one click

Handling Tough Questions

Question	Answer
"How accurate is extraction?"	10+ sanity checks (PAT < Revenue, GNPA > NNPA, etc.). Dual-track extraction provides cross-validation.
"What about scanned PDFs?"	PyMuPDF primary + OCR fallback for scanned/image-only pages.
"Is the ML model trained on real data?"	Yes, train_from_pdfs() extracts from real annual reports, augments with synthetic data centered on real distributions.
"How different from CIBIL?"	We analyze unstructured data (annual reports, news, legal) that bureau scores miss. Plus GST-Bank cross-verification for fraud.